Essays

|!|BCEEC0300000001|!|

Discuss how the study of fossils give evidence for evolution and the limitations of using fossils as evidence for evolution. (12 marks)

##

Fossils as evidence for evolution: (6)

- By studying fossils, we can learn about the variety and distribution of organisms in the past. (1)
- By arranging fossils of organisms with close resemblance according to their time of existence, we can observe how organisms changed over time. (1)
- A greater diversity of fossils is found in younger rocks than in older rocks. (1)
  This agrees with the hypothesis that organisms evolved from a common ancestor. (1)
- Fossils in younger rocks are more complex than those in older rocks. (1)
  This agrees with the hypothesis that organisms evolved from simple to complex forms. (1)

Limitations of using fossils as evidence for evolution: (Max. 3)

- The fossil record is incomplete. / There are missing links in the fossil record. (1)
- Some fossils are incomplete or damaged. (1)
- Some fossils are located at inaccessible areas. (1)
- Soft bodied organisms may not form fossils. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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|!|BCEEB0300000002|!| ★ **Cross topic**

Explain the structure of proteins in organisms, and how the study of proteins provide evidence for evolution. (12 marks)

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Structure of proteins in organisms: (5)

- The basic units that form protein molecules are amino acid. (1)
- Many amino acids are joined together, forming a polypeptide chain. (1)
- A protein molecule may consist of one or more polypeptide chains. (1)
- The amino acid sequence determines the shape (conformation) of a protein molecule. (1)
- Hydrogen bonds that form among amino acids cause polypeptide chains to coil and fold into a variety of shapes, such as helices, sheets and globular structures. (1)

How the study of proteins provide evidence for evolution: (4)

- All organisms use the same 20 types of amino acids to synthesize proteins as directed by the base sequences of DNA. (1)
- Certain essential proteins are commonly found in many different organisms. Their amino acid sequences may be slightly different as a result of mutations. (1)
- The amino acid sequences of essential proteins in different organisms can be compared to deduce the evolutionary relationship among them. (1)
- Scientists assume that the fewer the number of amino acid differences of a protein found in different organisms, the more closely related the organisms are. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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